

# Solutions — Variables in Python

## Module 1 | Python Programming Fundamentals

### Topic 1 of 3: Variables

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**Note for Instructors:** This file contains solutions to the 10 student practice tasks from the Variables Lab Manual. Distribute only after students have attempted the tasks independently.

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#### Solution 1 — Personal Variables

```
full_name = "Yousif Ahmed"
age = 21
city = "Karachi"

print(full_name)
print(age)
print(city)
```

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#### Solution 2 — Swap Values

```
a = 50
b = 80
print("Before:", a, b)

a, b = b, a
print("After:", a, b)
```

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#### Solution 3 — Counter Increment

```
counter = 0
counter = counter + 1
counter = counter + 1
counter = counter + 1
counter = counter + 1
counter = counter + 1
print(counter) # Output: 5
```

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#### Solution 4 — Memory Address with `id()`

```
x = 5
y = 5
print(id(x))
print(id(y))
print(id(x) == id(y)) # Likely True for small integers (Python caches them)
```

**Observation:** Python caches small integers (-5 to 256), so x and y point to the same object in memory. For larger numbers this would not be the case.

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## Solution 5 — Type Change with Reassignment

```
temperature = 36.6
print(temperature)          # 36.6

temperature = "Fever"
print(temperature)          # Fever
# Python allowed this – it's dynamically typed.
```

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## Solution 6 — Formatted Student Info

```
name = "Ali"
roll = 101
cgpa = 3.75

print(f"Student {name} (Roll: {roll}) has a CGPA of {cgpa}.")
```

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## Solution 7 — Multiple Assignment in One Line

```
mon, tue, wed, thu, fri = 1, 2, 3, 4, 5
print(mon, tue, wed, thu, fri)
```

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## Solution 8 — Shopkeeper Receipt

```
item = "Notebook"
price = 150
quantity = 4

total = price * quantity
print(f"Item: {item}")
print(f"Price per unit: Rs. {price}")
print(f"Quantity: {quantity}")
print(f"Total: Rs. {total}")
```

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## Solution 9 — Tracing Reassignment

```
message = "Hello!"
print(message)

message = "How are you?"
print(message)

message = "I am learning Python."
print(message)

message = "Variables are fun!"
print(message)
```

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## Solution 10 — Constants

```
COUNTRY = "Pakistan"  
CAPITAL = "Islamabad"  
POPULATION = 230_000_000
```

```
print(f"{COUNTRY} is a country in South Asia. Its capital is {CAPITAL} and it  
has a population of approximately {POPULATION} people.")
```